

DIGITAL WAITERS: INTRO TO REST APIs

Software Engineering Fundamentals — Practical Conceptual Lesson Guide

Subject: Computer Science & Web Engineering **Level:** Young Engineers & Tech Enthusiasts **Module:** Foundations of REST APIs

1. Understanding Real-Time App Data

When you open an app like Zomato or Swiggy, it instantly shows live restaurant statuses across the city. Modern applications do not store all this changing data locally on your smartphone; instead, they dynamically request fresh information from remote servers via an **API (Application Programming Interface)**.

2. The Digital Waiter Analogy

How an API Works in Real Life:

- **The Customer (Client / Phone):** You look at options and request specific data/dishes.
- **The Kitchen (Server / Database):** Holds the raw resources, data, and capabilities to fulfill requests.
- **The Waiter (The API):** Takes your order to the kitchen and delivers the response back to your table.

3. What is an Endpoint?

APIs organize their services using path routes called **endpoints**. Think of entering a department store like Big Bazaar: the *Dairy Counter* serves milk, while the *Electronics Counter* serves gadgets. Similarly:

- `/milk` or `/canteen/menu` fetches menu or item data.
- `/canteen/order` handles order submissions.

4. The 4 Basic API Action Verbs

HTTP Verb	Core Objective	Waiter Analogy
GET	Fetch / Read existing data safely	Asking the waiter to show you today's physical menu.
POST	Create / Submit new records	Placing a brand-new food order at the counter register.
PUT	Update / Modify existing data	Changing your order choices before the food is prepared.
DELETE	Erase / Remove records permanently	Canceling your food order completely.

5. JSON & The Request-Response Cycle

Computers exchange structured data using **JSON (JavaScript Object Notation)**—a lightweight text format based on key-value pairs:

```
{
  "restaurantName": "Canteen Express",
  "isOpen": true,
  "availableItems": ["Samosa", "Sandwich", "Juice"]
}
```

The Request-Response Journey: (1) Client builds request → (2) Waiter carries request over network → (3) Server processes & builds response → (4) Waiter brings data back → (5) App displays updated UI!

STUDENT WORKSHEET: REST API SYSTEM ARCHITECT

Design your own custom API architecture and map out HTTP verbs, endpoints, and JSON models!

Activity 1: School Canteen API Blueprint

Fill out the API Specification table for a School Canteen App system:

Task / Feature	Endpoint Path	HTTP Verb	Expected Payload / Behavior
View Daily Menu	/canteen/menu	GET	Returns array of available items & prices.
Place Lunch Order	/canteen/order	POST	Submits item selection; returns Order ID.
Update Seat Number			Modifies delivery seat location for active order.
Cancel Active Order			Erases order from kitchen queue.

Activity 2: Constructing JSON Data Models

Write a valid JSON payload snippet representing a student profile requesting lunch. Include key-value pairs for `studentName`, `grade`, `isVegetarian`, and an array `orderedItems`:

```
{
  "studentName": "_____",
  "grade": 7,
  "isVegetarian": false,
  "orderedItems": [
    {
      "id": 1,
      "name": "Chicken Sandwich",
      "price": 5.50
    },
    {
      "id": 2,
      "name": "Veg Burger",
      "price": 4.00
    }
  ]
}
```

Activity 3: HTTP Verb Mapping Challenge

Match each real-world app interaction to its appropriate HTTP verb (GET, POST, PUT, or DELETE):

App Scenario	Correct HTTP Verb
1. Reloading your Instagram feed to see new posts	
2. Publishing a new photo with a caption	
3. Changing your profile bio description	
4. Removing an old post from your gallery	

STUDENT PRACTICE ASSESSMENT

Foundations of REST APIs — Concept Review & Problem Solving

Instructions: Answer all questions in the space provided. No reference answers are included on this sheet.

Section A: Multiple Choice Questions

Q1. In the "Digital Waiter" analogy, what component does the Waiter represent?

- [A] The Database Server
- [B] The Client / Smartphone User
- [C] The Application Programming Interface (API)
- [D] The Physical Wi-Fi Router

Q2. Which HTTP verb is used when an application only needs to read or fetch data without making any changes on the server?

- [A] POST
- [B] GET
- [C] PUT
- [D] DELETE

Q3. What does JSON stand for in web development?

- [A] Java Standard Output Network
- [B] JavaScript Object Notation
- [C] Joint System Operations Node
- [D] Judicial Server Open Syntax

Q4. Which of the following best defines an API Endpoint?

- [A] The physical power cable connecting a server to the wall
- [B] A structured path route (e.g. /canteen/menu) that targets a specific server resource
- [C] The final screen shown when an application crashes
- [D] A password used to log into an administrator dashboard

Section B: Short Answer & Conceptual Mastery

Q5. Why can't a mobile food delivery app keep all restaurant status data stored directly inside the app download file? Explain what would happen if it did.

Q6. Trace the 5 steps of the Request-Response Cycle when a user taps "Refresh Menu" on their phone screen.

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Section C: Real-World Scenario Problem

Q7. Scenario: You are building a Ride-Hailing App (like Uber or Ola). A passenger opens the app, views available drivers nearby, requests a ride, updates their drop-off location, and later cancels the ride.

- (a) Identify which HTTP verb (GET, POST, PUT, DELETE) is used for each step of the journey.
- (b) Propose a clean endpoint path for requesting a new ride.

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